



TP拟合海洋剖面实验进展

上层200米深度测试

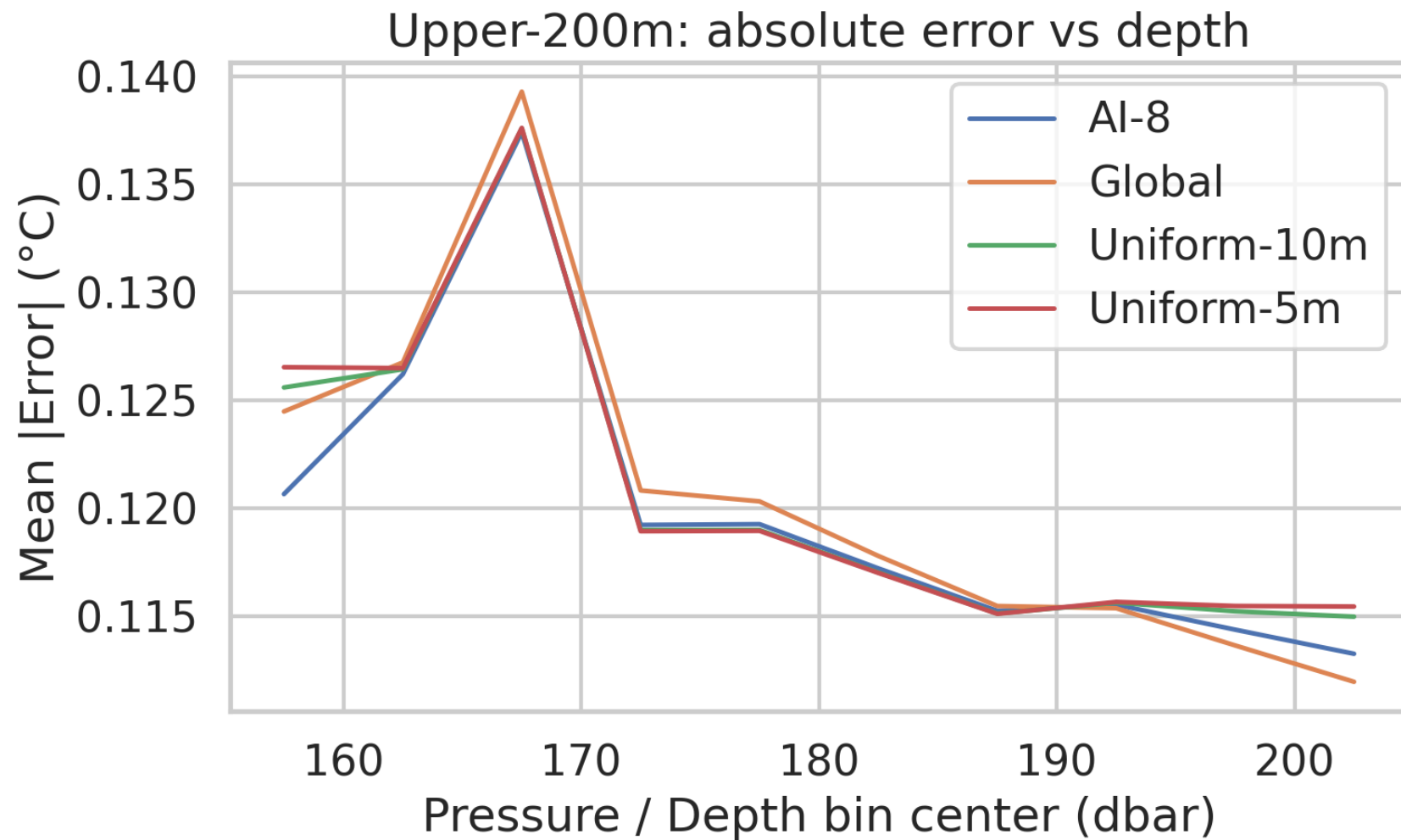
在上层200米深度，用AI-智选的8个测量点，同每10米测一个点，每5米测一个点，及global 比较准确性

- Global (全量拟合, 上限对照; 评估仅看 0–200 m)
- AI-8 (Upper-200 D-optimal 智选 8 点)
- Uniform-10m (每 10 m 一点, 避开 0, 包含 1 dbar)
- Uniform-5m (每 5 m 一点, 避开 0, 包含 1 dbar)

四法的总体误差非常接近，AI-8 与 Uniform-10m/5m 的 MAE/RMSE/MaxAbs 差异在千分量级：

- Global: `MAE=0.3602`, `RMSE=0.4808`, `MaxAbs=1.500`, `Std=0.4807`
- AI-8: `MAE=0.3586`, `RMSE=0.4802`, `MaxAbs=1.486`, `Std=0.4802`
- Uniform-10m: `MAE=0.3589`, `RMSE=0.4803`, `MaxAbs=1.493`, `Std=0.4803`
- Uniform-5m: `MAE=0.3591`, `RMSE=0.4804`, `MaxAbs=1.494`, `Std=0.4804`

上层200米深度测试



系数舍入敏感性实验

8项多项式，准确度到了小数点后第三位。试试看，再增加几项多项式，可以到第四位？

将闭式方程 (Mixed B / Eq. B.1) 的 8 个常数系数分别保留到小数点后三位与四位，评估对误差指标的影响。
评估域：上层 200 m ($P \in [1, 200]$)，QC=1；与老师关注域一致，同时避开 $\ln P$ 的 $P=0$ 奇点。

系数版本- Full (论文原始精度)：`[-356.873, 0.0125047, 62.2025, -3.35576, 20.0563, 14.8832, 10011.3, 7576.48]`

- Round-3：按小数点后三位四舍五入
- - Round-4：按小数点后四位四舍五入
- 结果摘要 (上层 200 m) - Full：`MAE=0.36021`，`RMSE=0.48080`，`MaxAbs=1.49990`，`Std=0.48072` -
Round-3：`MAE=0.37770`，`RMSE=0.48931`，`MaxAbs=1.40726`，`Std=0.48053`，`ΔMAE=+0.01749`，
`ΔRMSE=+0.00851`
- - Round-4：`MAE=0.36005`，`RMSE=0.48078`，`MaxAbs=1.50140`，`Std=0.48073`，`ΔMAE=-0.00016`，
`ΔRMSE≈-0.00002`
- “将 Mixed B 系数保留到 3 位小数会引入约 `+0.0175` 的 MAE 增量与 `+0.0085` 的 RMSE 增量；提升到 4 位小数后增量近乎为 0。因此部署建议至少保留 4 位小数 (或对小系数保留 4 位、对大系数保留 3 位)。”

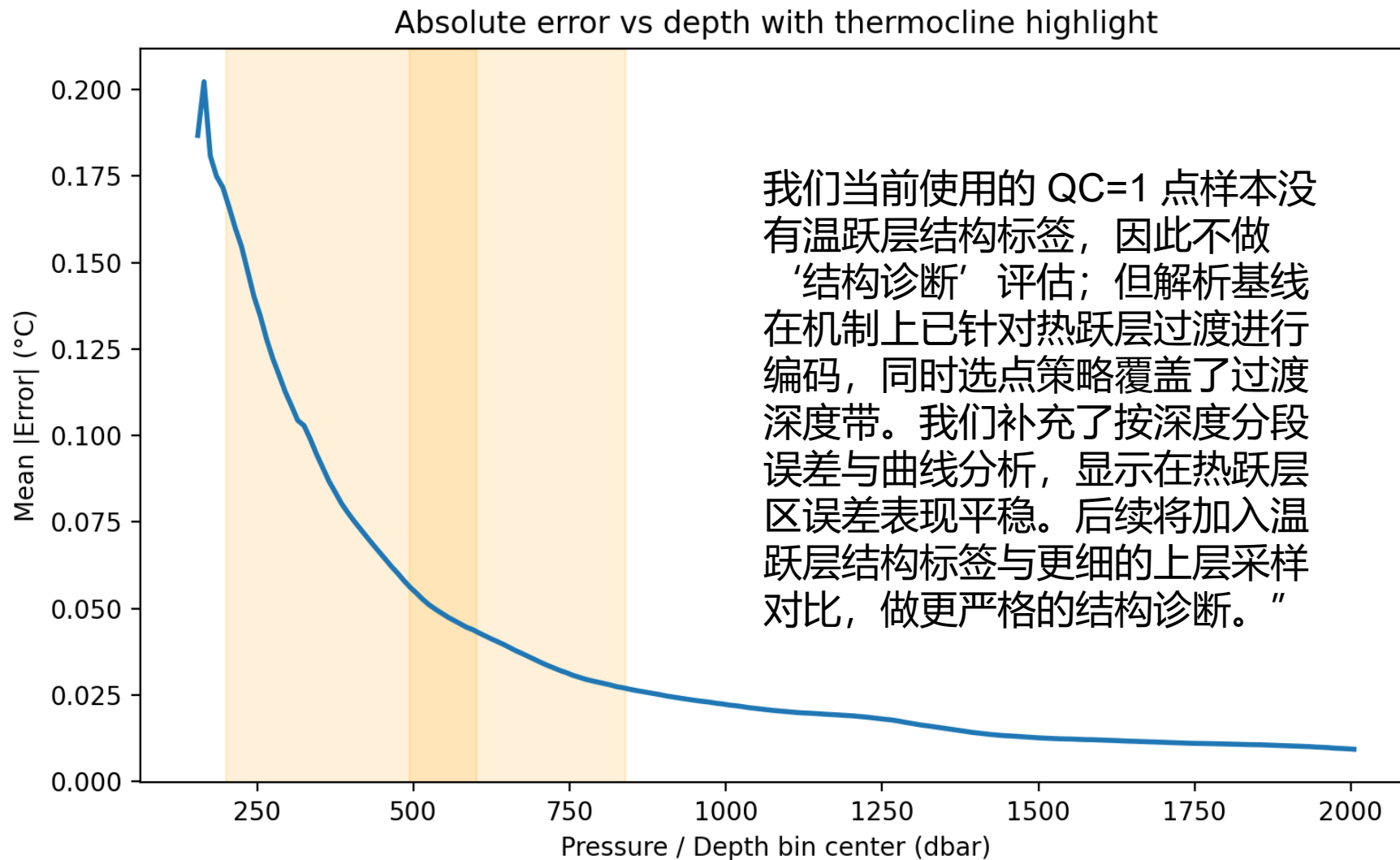
整体 T-P 关系的可解释重建能力

范围界定- 数据性质：使用 QC=1 的大规模 T-P 点样本；不包含温跃层结构性标签（如顶/底、强度、MLD），且上层垂向采样密度受设备与任务影响可能不均匀。因此本文评估的是整体 T-P 关系的可解释重建能力，而非“温跃层结构诊断”。

模型机制：解析基线 Mixed B 在基函数设计上显式考虑了混合层—温跃层过渡与中层曲率（指数项、sqrt/log）以及深海尾部（倒数项），以避免纯多项式在高曲率区域的振荡与大残差

影响评估（分段误差 + 深度曲线） - 表：`tables/thermocline\_segment\_errors.csv` - `0-200`：MAE≈0.375, RMSE≈0.412, MaxAbs≈1.500 - `200-600`：MAE≈0.171, RMSE≈0.206, MaxAbs≈1.559（热跃层常见带） - `493-838 (transition)`：MAE≈0.076, RMSE≈0.086, MaxAbs≈0.532（文中选点覆盖的过渡带） - `600-2000`：MAE≈0.0366, RMSE≈0.0458, MaxAbs≈0.428

最大绝对误差的指标



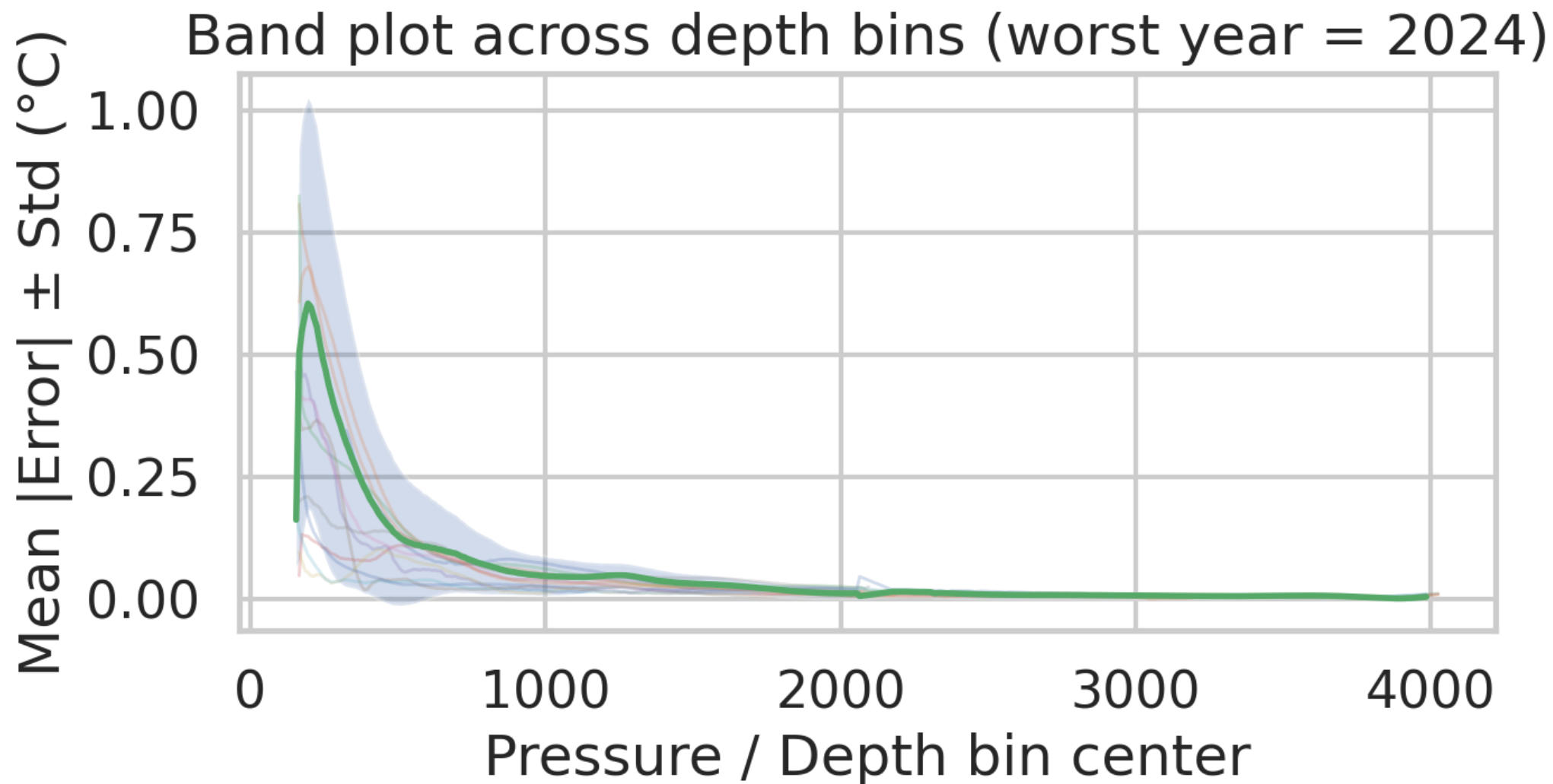
年份区分的误差

同一个 Global 方程（固定系数）在不同年份数据上评估，该年所有样本的最大绝对误差是否达到 0.1–0.2°C？ ”

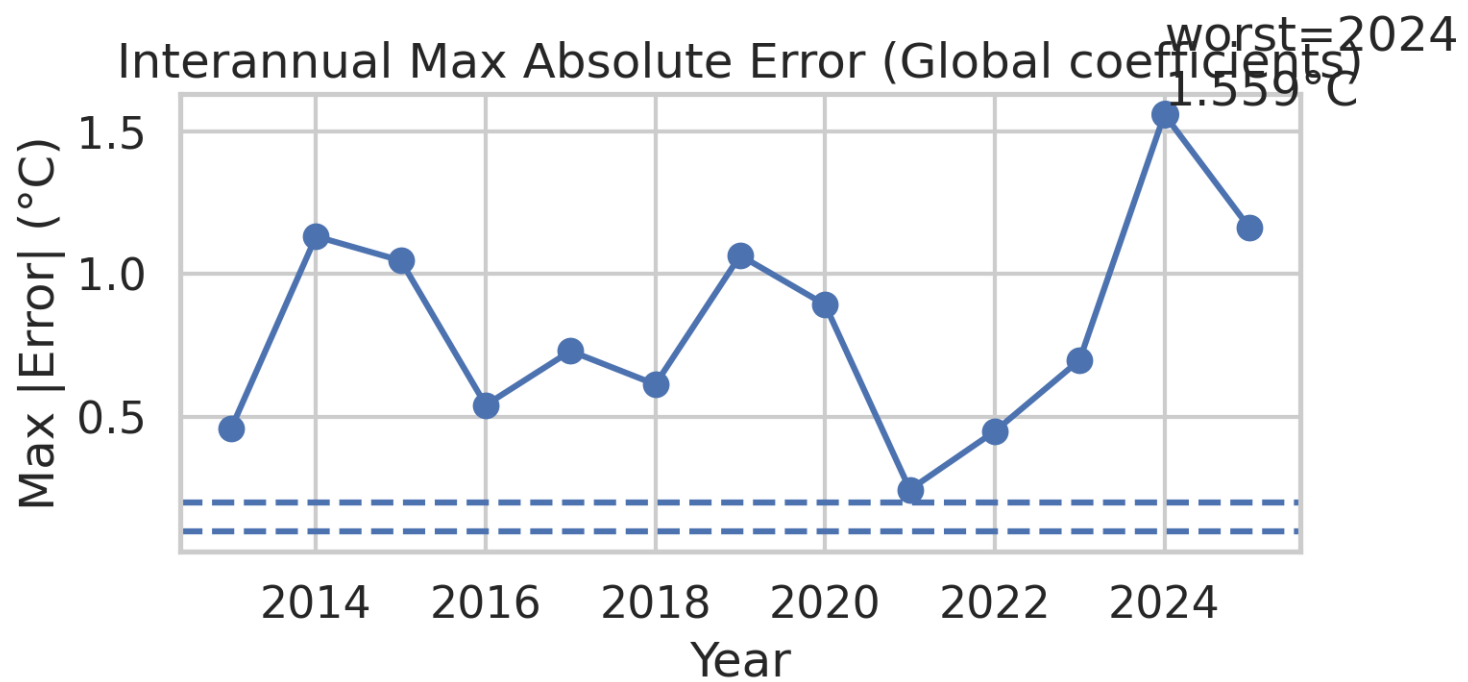
最差年份：`2024` 年 - `MaxAbsError = 1.559°C` - `P99AbsError = 0.938°C` - `Std = 0.152°C` - `N = 4,163,923`

- 代表性 “好年” ： - `2021` 年：`MaxAbsError = 0.245°C` , `P99AbsError = 0.116°C` , `Std = 0.027°C` - `2022` 年：`MaxAbsError = 0.449°C` , `P99AbsError = 0.107°C` , `Std ≈ 0.029°C` - 对 “0.1–0.2°C”问题的回应：从图1与表可见，许多年 `P99AbsError` 落在 ~0.1°C 量级，但部分年份的 “极值” 明显超过 0.2°C；最差年已在图中明确标注。

年份区分的误差



年份区分的误差

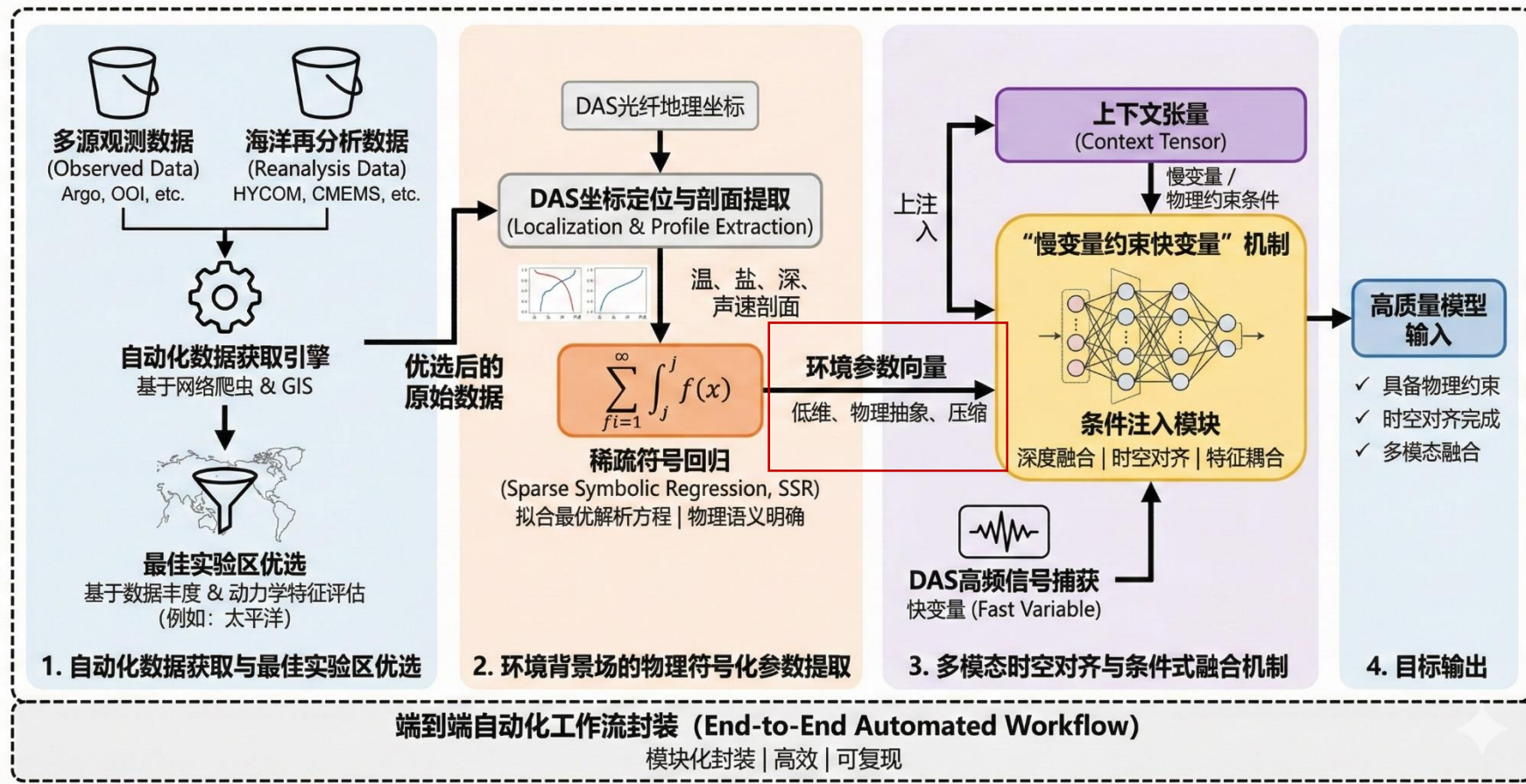




开题报告合适数据调研

数据处理流程总结

图 1：基于物理参数化驱动的端到端海洋多模态数据融合与标准化工程体系



点式传感数据
拟合剖面
形成物理参数
与光纤数据融合
进行模型识别工作的
数据基础

高精度剖面拟合以及物理参数输出工作已在最新的研究文章中得到验证

目前锁定合适的实验区域

REGIONAL CABLED & ENDURANCE ARRAYS

SUMMIT 1

- Low Voltage Node (LV01B)
- Low Powered J-Box (LJ01B)
- Tidal Seafloor Pressure
- Low Frequency Acoustic Receiver (Hydrophone)
- 3 Short-Period Ocean Bottom Seismometers
- Broadband Ocean Bottom Seismometer

SOUTHERN HYDRATE RIDGE 780 M

SUMMIT 2

- Med. Powered J-Box (MJ01B)
- Digital Still Camera
- Acoustic Doppler Current Profiler
- Mass Spectrometer
- Osmosis-Based Water Sampler
- 2 Benthic Fluid Flow

INSHORE 25 M

- Surface Mooring (CE01ISSM)
- Surface Piercing Profiler Mooring (CE01ISSP)

SHELF 80 M

- Surface Mooring (CE02SHSM)
- Surface Piercing Profiler Mooring (CE02SHSP)
- Cabled Benthic Experiment Package (CE02SHBP)
- Primary Node (PN1D)
- Med. Powered J-Box (MJ01C)
- Digital Still Camera
- Broadband Acoustic Receiver (Hydrophone)
- Bio-acoustic Sonar (Coastal)

SLOPE BASE 2900 M

- Primary Node (PN1A)
- Medium Powered J-Box (MJ01A)
- Cabled Shallow Profiler Mooring (RS01SBPS)
- Cabled Deep Profiler Mooring (RS01SBPD)
- Broadband Ocean Bottom Seismometer
- Low Frequency Acoustic Receiver (Hydrophone)
- Tidal Seafloor Pressure
- 3-D Single Point Velocity Meter

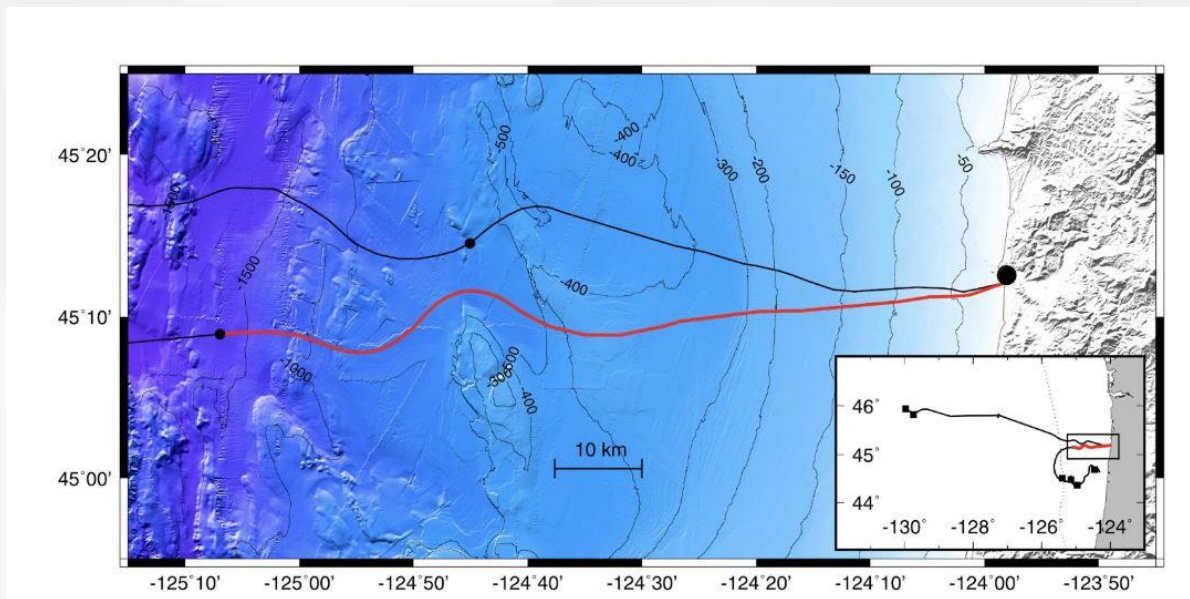
OFFSHORE 600 M

- Cabled Shallow Profiler Mooring (CE04OSPS)
- Cabled Deep Profiler Mooring (CE04OSPD)
- Cabled Benthic Experiment Package (CE04OSBP)
- Surface Mooring (CE04OSSM)
- Primary Node (PN1C)

美国俄勒冈州纽波特 (Newport, OR) 外海, 从近岸浅水区延伸至深海盆地的海底观测网络

既有光纤 (作为传输和传感基础设施), 也有 CTD (作为核心观测仪器)。

目前锁定合适的实验区域



美国 OOI (Ocean Observatories Initiative) Regional Cabled Array (RCA) 项目中。华盛顿大学 (University of Washington) 的团队在俄勒冈海岸 (Oregon) 铺设的海底光缆上进行了专门的DAS实验，而该海域本身就布满了OOI的常驻CTD传感器。





论文进度

论文进度

文件 编辑 插入 查看 格式 帮助

文件树

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Fig1.png

Fig2.png

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Fig4.png

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Introduction

Related Work

Evolution from Statistical Inference to D...

State-of-the-Art in Ocean Data Quality C...

Challenges and Architectural Advances I...

Methodology

Hybrid Basis Function Design

Automatic Basis Function Search Mecha...

Residual Neural Network Structure and ...

Methodology Summary

Experimental Setup

Data Acquisition and Preprocessing

Experiment I: Comparative Analysis of G...

Seasonal Partitioning Equation-Inferenc...

Depth-Stratified Fitting Experiment

Rapid Coefficient Inference Experiment ...

Baseline Model Performance Comparis...

Temporal Robustness and Multi-Scale Er...

源代码编辑器 可视化编辑器

Normal text

sn-article.tex > Experimental Setup

235 \item{\textit{complementary synergy}}: The analytical component models low-frequency trends, while the residual network compensates for high-frequency details, achieving multi-scale collaboration.

236 \end{itemize}

237

238 Overall, the residual network serves as an \emph{adaptive corrector} within the proposed framework. It does not replace the physical model but dynamically fine-tunes the deficiencies of the analytical part. The analytical term $S(P; \mathbf{w})$ provides a deterministic physical baseline, whereas $S(P, z; \boldsymbol{\theta})$ learns nonlinear features from data, enabling the model to achieve both thermodynamic consistency and high-resolution predictive accuracy.

239

240 \subsection{Methodology Summary}

241 In summary, the proposed hybrid modeling framework integrates physically interpretable analytical modeling with data-driven residual learning to achieve accurate, stable, and physically consistent temperature-pressure \textit{prediction} in deep-sea environments. As visually summarized in Figure \ref{fig:hybrid_framework}, the framework operates through a synergistic "Structure-Residual" decomposition. The analytical component $S(P; \mathbf{w})$ constructs a thermodynamically meaningful baseline through hybrid basis functions, ensuring smoothness, \textit{interpretability}, and stability across varying pressure ranges. The automatic Top-Ks search mechanism further enhances adaptability by optimizing functional parameters in a data-driven manner, eliminating human bias and improving generalization across diverse marine conditions. On top of this foundation, the residual neural network $S(P, z; \boldsymbol{\theta})$ complements the analytical model by capturing high-frequency and nonlinear components using Fourier feature mapping and gradient-enhanced feedback. This cooperative architecture follows the principle of "low-frequency modeled by physics, high-frequency refined by learning", unifying physical interpretability with data adaptivity. As a result, the framework not only accelerates convergence and enhances robustness but also provides a physically grounded and computationally efficient pathway for complex thermodynamic modeling in deep-sea systems.

242

243 \begin{figure}[htbp]

244 \centering

245 \includegraphics[width=1.0\textwidth]{fig5.png}

246 \caption{Schematic of the Unified Hybrid Framework (Ocean-TP).}

247 The framework decomposes the modeling task into two complementary pathways: (Left) An "Analytical Trend Module" constructs a physically interpretable baseline using hybrid basis functions to capture dominant low-frequency trends; (Right) A "Residual Learning Module" employs a gradient-enhanced neural network to resolve high-frequency nonlinear perturbations. The final prediction integrates both components to ensure thermodynamic consistency and high precision.

248 \label{fig:hybrid_framework}

249 \end{figure}

250

251 % ===== Experiments =====

252

253 \section{Experimental Setup}

254

255 \subsection{Data Acquisition and Preprocessing}

256

257 The experimental dataset was constructed from the multi-source \textit{Ocean Observatories Initiative (OOI)} global array. To isolate a representative deep-sea environment with complex stratification features, we implemented an \textit{automated geographic screening algorithm} to traverse the OOI database, identifying and extracting high-density cast data specifically from the \textit{Gulf of Alaska} region. The raw data from these heterogeneous sources underwent a rigorous fusion and quality control pipeline via the proposed \textit{Physics-based Data Deblurring and Cleaning (DDC)} algorithm. By enforcing TEOS-10 stability criteria and removing invalid entries, we fused the multi-source inputs into a unified, high-quality experimental dataset containing \textit{27,107,765} valid samples ($Q \geq 1$). Finally, all negative depth values were converted to positive pressure SPS (dbar) to ensure a consistent coordinate system for modeling.

258

259 \subsection{Experiment I: Comparative Analysis of Global Equation Structures}

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261 To identify the optimal analytical form for the global ocean temperature-pressure relationship, we designed a systematic equation-structure comparison experiment. The objective was to assess the expressive capacity and stability of different analytical formulations in representing the vertical distribution of seawater temperature, thereby identifying an interpretable and generalizable functional form. Eight candidate equation structures were constructed to model the global temperature profile. These candidates cover a spectrum of complexity, including polynomial expansions, normalized rational functions, and hybrid mixtures involving logarithmic ($S(\ln P)$), exponential ($S(\exp(-P/\beta))$), and hyperbolic tangent ($S(\tanh P)$) terms. Each structure reflects distinct physical hypotheses regarding vertical stratification, such as the rapid decay in the shallow mixed layer, the transitional curvature of the thermocline, and the asymptotic stability of deep-water zones. Each candidate global model was trained within a unified \textit>PNN} framework to ensure a fair comparison. The input variable was pressure SP (or normalized depth z), and the output was the predicted temperature $S(\text{hat{T})$. The total loss function $S(\text{mathcal{L}})$ combined data fidelity with physical residual terms:

262 \begin{equation}

263 \mathcal{L} = \lambda_{\text{data}} d_{\text{L}} + \lambda_{\text{thrm}} MSE(\hat{T}, T) + \lambda_{\text{data}} p_{\text{L}} \left\| \frac{\partial \hat{T}}{\partial z} \right\| + \lambda_{\text{thrm}} \left(\frac{\partial \hat{T}}{\partial z} \right)^2

264 \end{equation}

265 where the first term enforces data fitting accuracy and the second imposes the physical gradient constraint to maintain thermodynamic consistency. The adaptive weighting coefficients λ_{data} and λ_{thrm} were dynamically adjusted according to the gradient norm ratio to balance learning between data-driven and physics-driven components. To isolate the performance impact of the equation structure, all candidate models shared an identical neural network architecture consisting of 5 fully connected layers with 64 hidden units each, utilizing \textit{Leaky ReLU} activation functions. A uniform training strategy was applied, incorporating \textit>warmup} cosine learning rate scheduling, early stopping, and Latin Hypercube Sampling (LHS) for collocation points, ensuring consistent experimental conditions across all tests.

结果和讨论还差点

Intro还需要润色

参考文献不够

图还没画好